

Is Ethereum Incentive-Compatible?

IC3 · 2026

Ulysse Pavloff

Yackolley Amoussou-Guenou · Sara Tucci-Piergiovanni

University of Bern





Motivation

Is following Ethereum's fork-choice rule a Nash equilibrium?

Obey the protocol

- Propose on the fork-choice head.
- Attest to the fork-choice head.
- Receive the standard rewards.

VS

Deviate for more

- Fork out previous blocks.
- Re-collect available fees and attestations.
- Still try to become canonical.

Ethereum assumes protocol-following behavior, but validators are rational payoff maximizers.

Proposer side

A proposer can gain rewards by:

- including more transactions,
- including more attestations,

**The older the parent,
the bigger the harvest.**

Block B_k

Header info

Included attestations:



Included transactions:

tx_1 : 0x8ac...91f

tx_2 : 0x3be...42a

tx_3 : 0x71d...e09

...

Attester's rewards

Timeliness	Next slot	≤ 5 slots	≤ 64 slots
Incorrect vote	$20x/27$	$20x/27$	$6x/27$
Correct vote	x	$20x/27$	$6x/27$

Next-slot inclusion: correctness matters. A correct head vote earns x instead of $20x/27$.

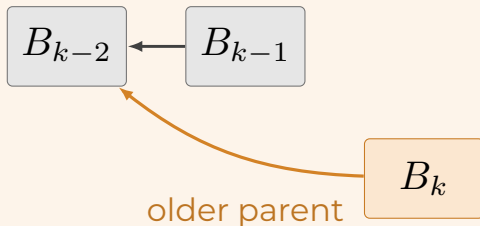
Later inclusion: correct and incorrect votes pay the same. This creates room for strategic anticipation.

An attester may care more about the next proposer than about the current head.

Two seeds of deviation

1. Proposer side

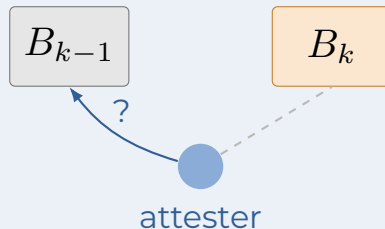
A proposer can sometimes build on an older block and still make its new block the head.



Goal: harvest more rewards.

2. Attester side

An attester's best reward depends on whether its vote matches the next canonical proposer.



Goal: maximize attestation reward by aligning with the next proposer.

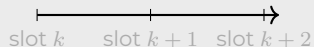


Ethereum in a nutshell

Ethereum PoS: the rules that matter

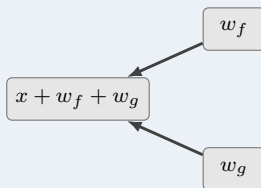
Slots

Each slot has one proposer and a attestations.



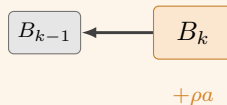
Fork choice

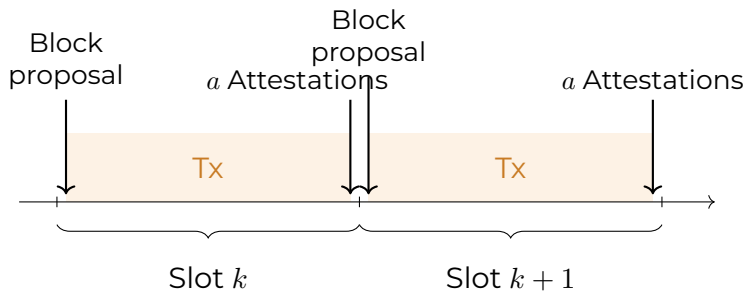
The head is selected by heaviest subtree of attestation weight.



Proposer boost

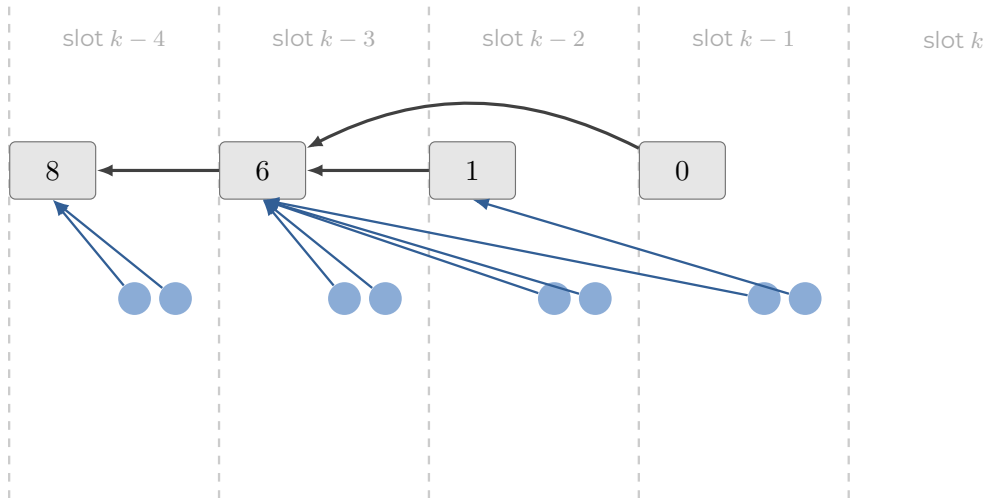
A timely block gets temporary weight $+\rho a$.





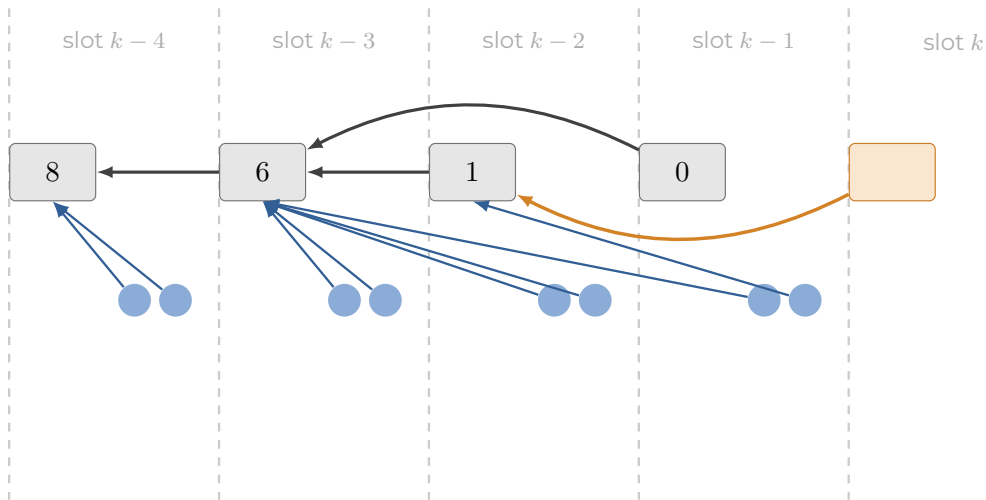
The game abstracts each round as block proposal, transaction generation, then simultaneous attestation broadcast.

Fork Choice Rule



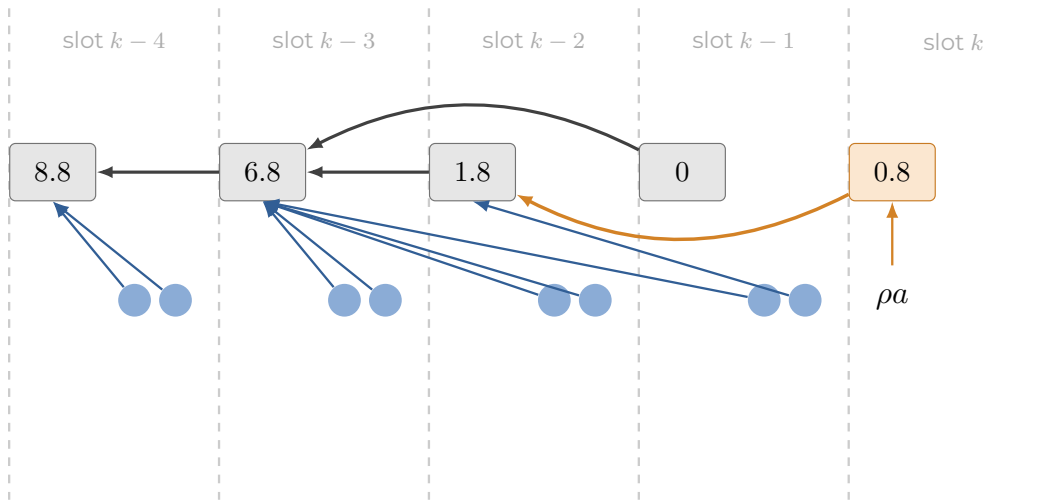
Before slot k , fork choice selects the head using the current attestation weights.

Fork Choice Rule



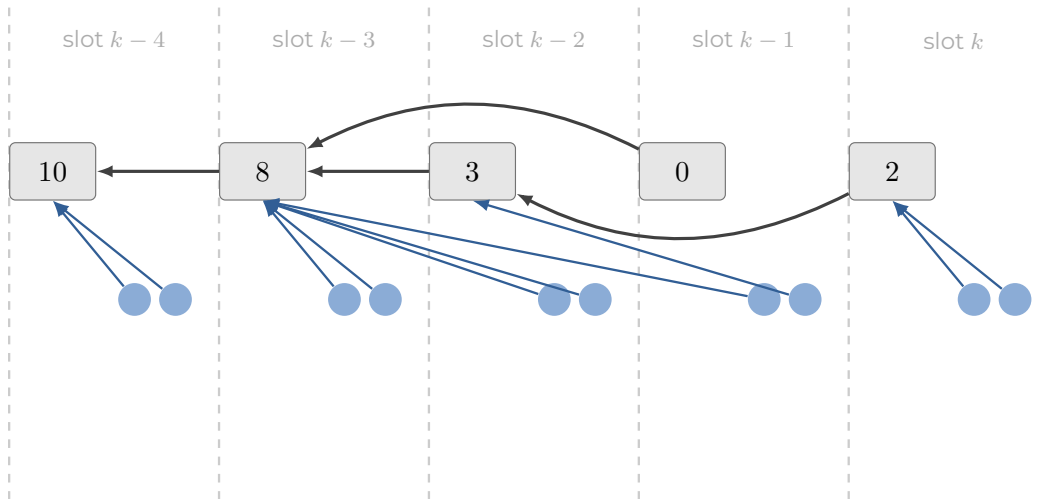
New block attach itself to the block designated by the FCR.

Proposer Boost: ρa



During slot k , the timely block has artificial weight $\rho a = 0.8$.

Proposer Boost: ρa



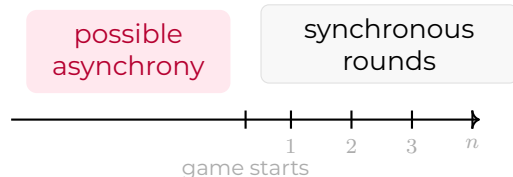
At the end of the slot, the boost is cleared and replaced by the two real attestations for B_k .



Model

Model & game

- Players are rational utility maximizers.
- Each round: one proposer and a attestors. (n rounds)
- The game is synchronous, but the initial state may come from a prior asynchronous period.
- No bribery, no collusion, no Byzantine behavior in this analysis. [1]

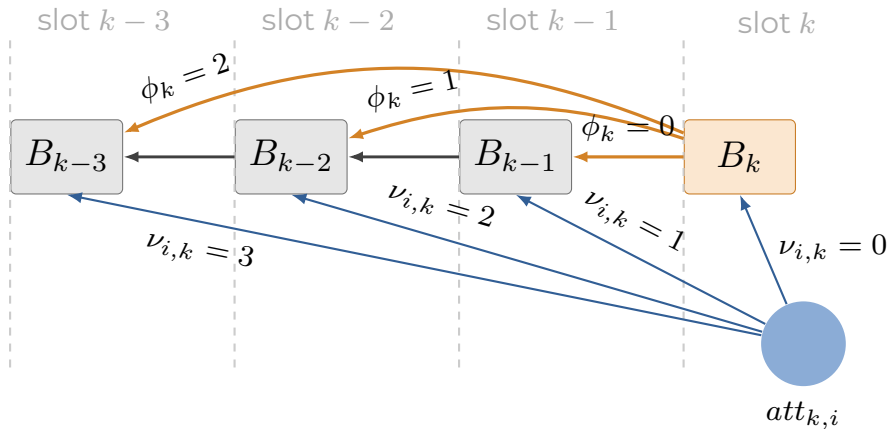


The key question is whether the prescribed strategy profile is a Nash equilibrium after such an initial state.



Actions

Actions: choosing where to build and where to vote



Proposer action: parent choice ϕ_k .

Attester action: block vote $\nu_{i,k}$.

Attester payoff. For attester (i, k) :

$$u_{i,k}(\sigma) = \begin{cases} x & \text{if the vote matches the next canonical proposer,} \\ 20x/27 & \text{if included within 5 slots,} \\ 6x/27 & \text{if included within 64 slots,} \\ 0 & \text{otherwise.} \end{cases}$$

Proposer payoff. For proposer $(0, k)$:

$$u_{0,k}(\sigma) = \chi_k \sum_{j=k-\phi_k}^k \left(f_{j-1} + \frac{1}{7} \sum_{i=1}^a u_{i,j}(\sigma) \right).$$

A proposer wants an old parent for more rewards, but **only if its block remains canonical.**



Strategies

Two strategy families

Obedient

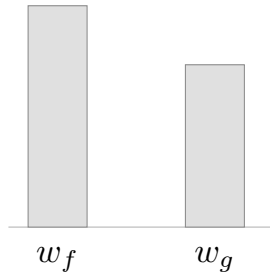
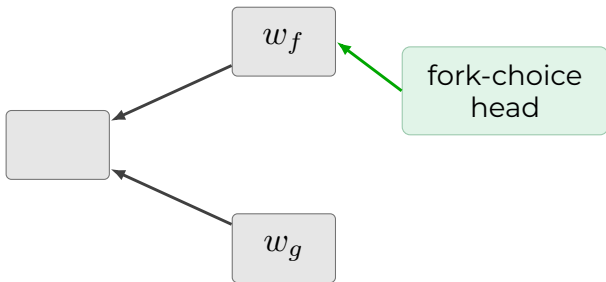
- Proposer: build on the fork-choice head.
- Attester: attest to the fork-choice head, including proposer boost.
- This is the prescribed protocol.

Cunning

- Proposer: choose the oldest parent that still makes its block head.
- Attester: vote for what the next cunning proposer will extend.
- Cunning may coincide with obedient when no deviation is profitable.

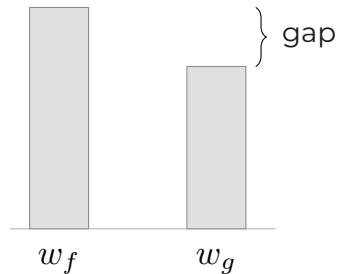
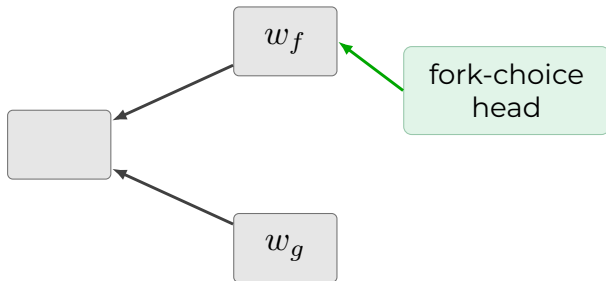
So the analysis asks: when does the cunning strategy actually differ from the obedient one?

The cunning condition



Two competing branches emerge after asynchrony — fork choice keeps the heavier one.

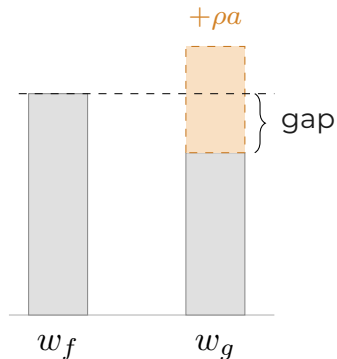
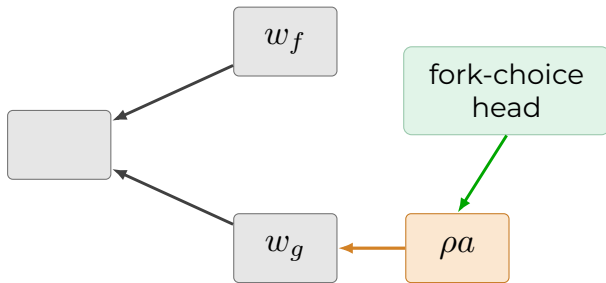
The cunning condition



How large is the weight gap between the two branches?

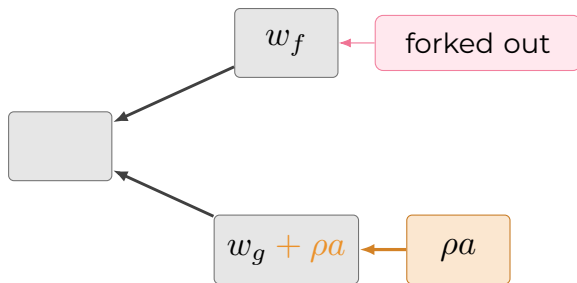
The cunning condition

$$w_g + \rho a \geq w_f$$

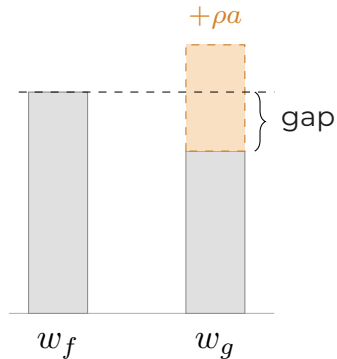


The proposer boost adds ρa to the lighter branch. If it covers the gap, the lighter branch is tipped to the top.

The cunning condition

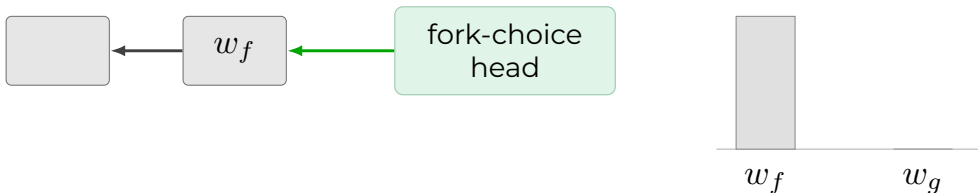


$$w_g + \rho a \geq w_f$$



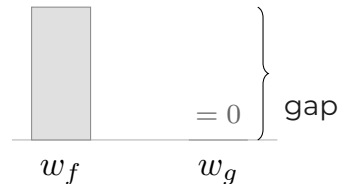
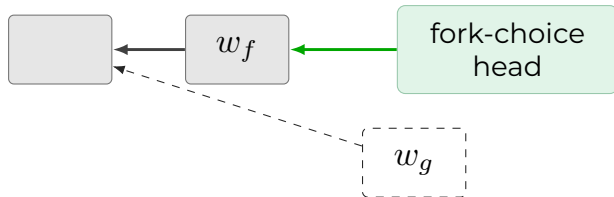
The cunning proposer reorgs the heavier branch and harvests its rewards.

The cunning condition: even without a pre-existing fork



A pre-existing fork is not necessary: fork choice simply keeps the current head.

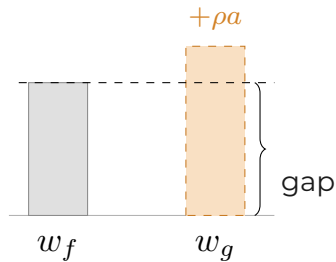
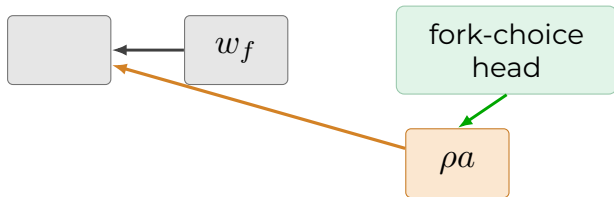
The cunning condition: even without a pre-existing fork



Here the competing branch has weight $w_g = 0$, so the condition becomes
$$w_f \leq \rho a.$$

The cunning condition: even without a pre-existing fork

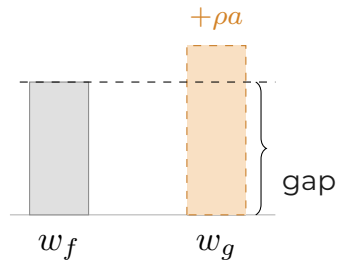
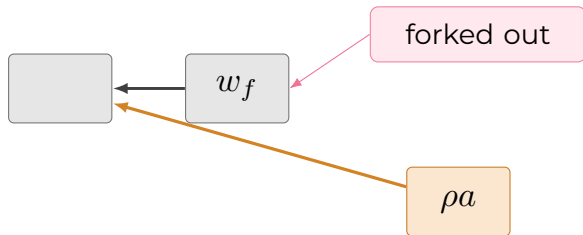
$$w_g + \rho a \geq w_f$$



The proposer creates a new branch and the boost may be enough to make it the new head.

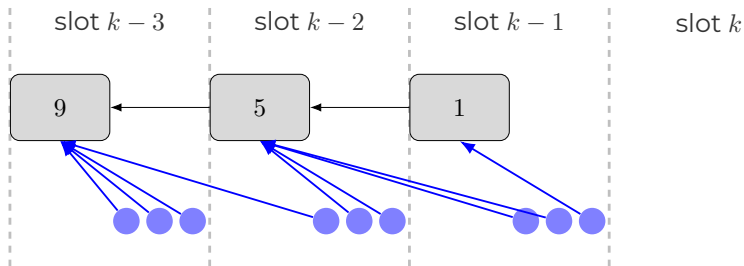
The cunning condition: even without a pre-existing fork

$$\rho a \geq w_f$$



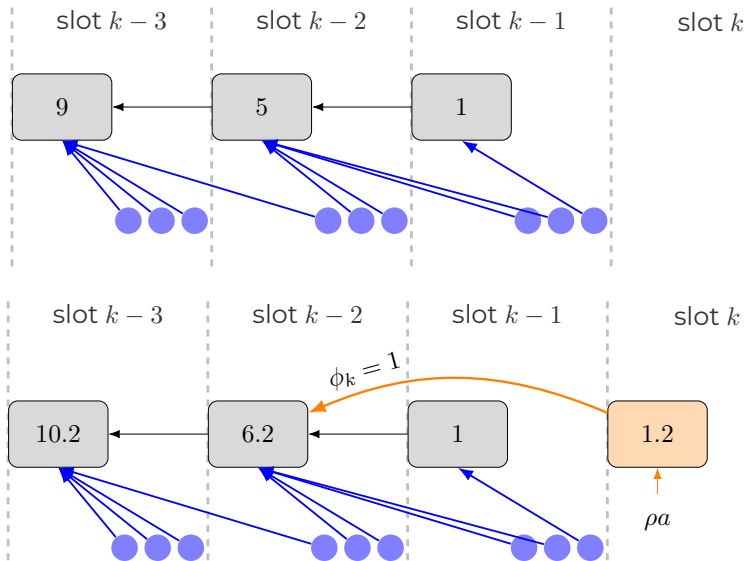
So the cunning condition also explains *fork creation*, not only choosing among existing forks.

Cunning condition: concrete example



$$\rho = 0.4 \text{ and } a = 3.$$

Cunning condition: concrete example



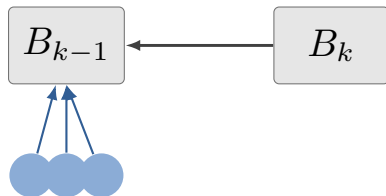


Results

RESULT 1 / 3: Fully synchronous $\Rightarrow$ obedience is stable

slot $k - 1$

slot k



a votes

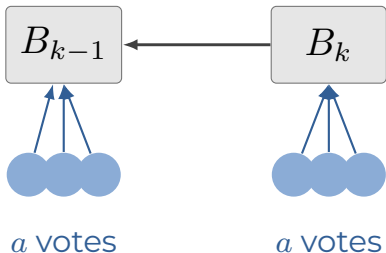
In synchrony, the fork-choice head receives the full committee support.

RESULT 1 / 3: Fully synchronous $\Rightarrow$ obedience is stable

slot $k - 1$

slot k

$$w_f - w_g \geq a$$

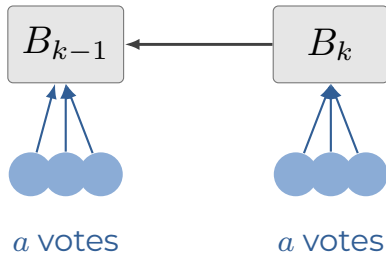


This creates a gap of at least one full committee weight.
The proposer boost is smaller than that gap. So strategic forking cannot succeed.

RESULT 1 / 3: Fully synchronous $\Rightarrow$ obedience is stable

slot $k - 1$

slot k



$$w_f - w_g \geq a$$

$$a > \rho a$$

since $\rho < 1$

Every slot adds a full committee of votes to the head. The gap becomes at least a , and $a > \rho a$. So the cunning condition never holds.

Obedience is a Nash equilibrium.

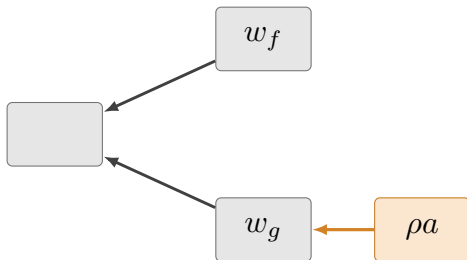


Result 2

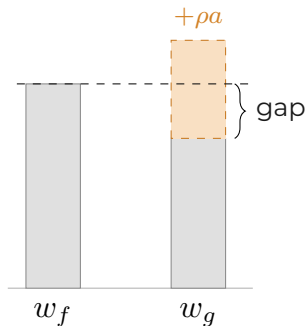
RESULT 2 / 3: After asynchrony

Asynchrony can leave uneven weights. Some branches may have less than a full committee of attestations. If the cunning conditions holds: $w_g + \rho a \geq w_f$, forks can occur.

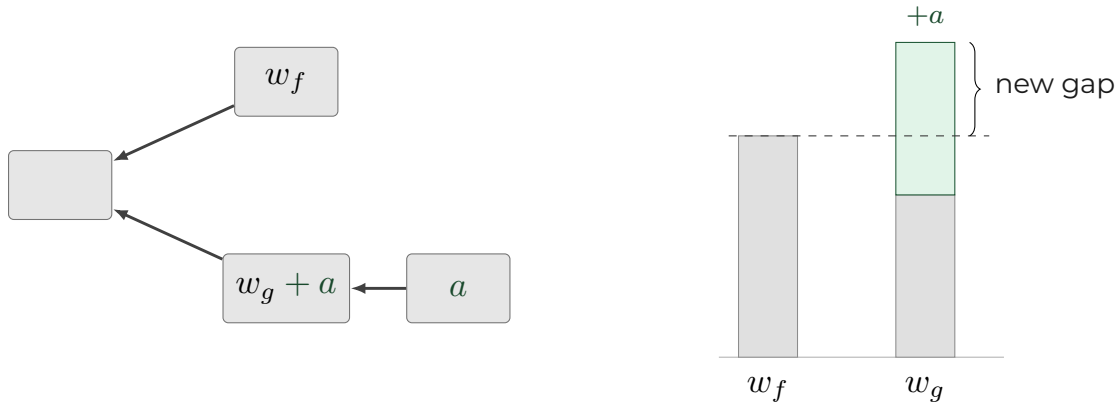
Obedience is not a Nash equilibrium.



$$w_g + \rho a \geq w_f$$

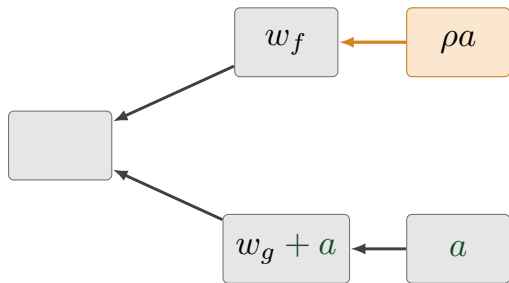


After one successful fork

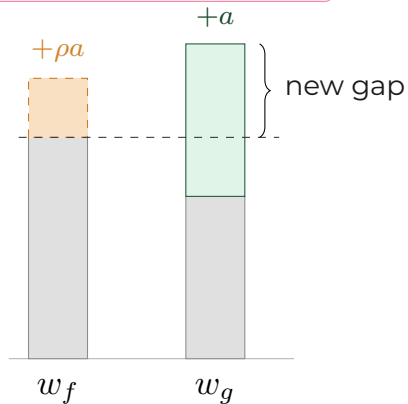


A successful cunning fork has occurred: the bottom branch is now the fork-choice head.

After one successful fork

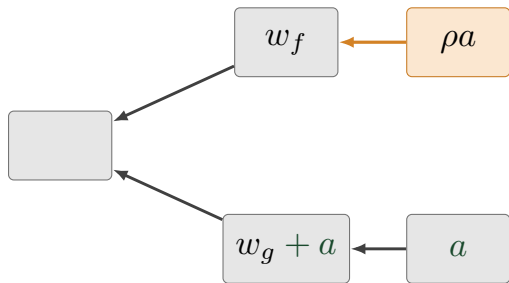


$$w_f + \rho a \geq w_g + a$$

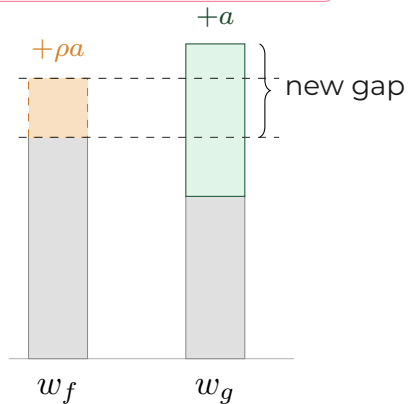


New cunning condition impossible impossible if $\rho < 1/2$.

After one successful fork

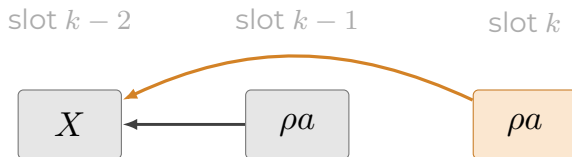


$$w_f + \rho a \geq w_g + a$$



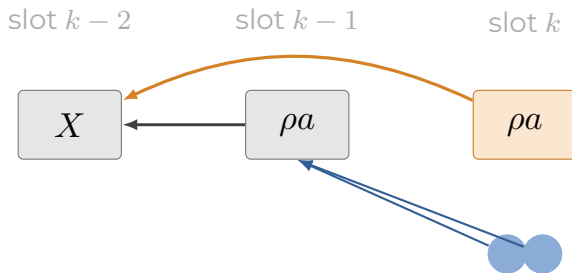
The next cunning condition is $(w_g + a) - w_f \leq \rho a. \Rightarrow \rho \geq 1/2$.

The surprise: a bigger proposer boost can help



If $\rho \geq 1/2$, the boost can make consecutive strategic forks technically possible.

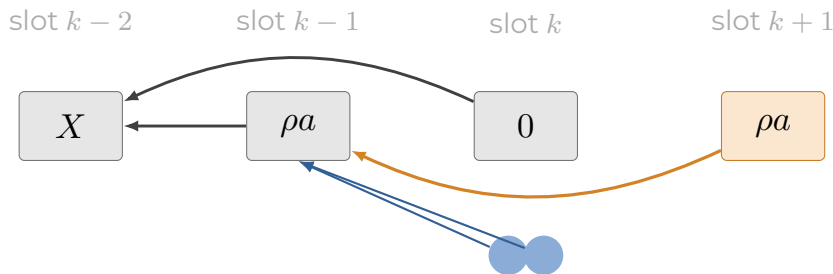
The surprise: a bigger proposer boost can help



If $\rho \geq 1/2$, the boost can make consecutive strategic forks technically possible.

But if attesters are also cunning, they become a credible deterrent: they do not support the lonely fork.

The surprise: a bigger proposer boost can help



If $\rho \geq 1/2$, the boost can make consecutive strategic forks technically possible.

But if attesters are also cunning, they become a credible deterrent: they do not support the lonely fork.

Larger boost $\Rightarrow$ fewer profitable forks.



Conclusion

Theorem : Eventual Incentive Compatibility

Theorem

In all Nash equilibria, there exists a round after which all validators follow the protocol.

Why it holds.

The cunning condition can hold for only a bounded number of rounds: transaction fees are positive and cannot decrease at every step indefinitely.

What it means.

Without any protocol change, Ethereum's fork-choice rule is eventually incentive-compatible under rational play.



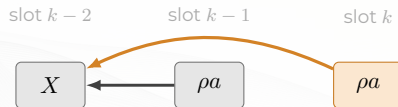
Thank you

for your attention

Incentive Compatibility of Ethereum's Proof-of-Stake Consensus Protocol

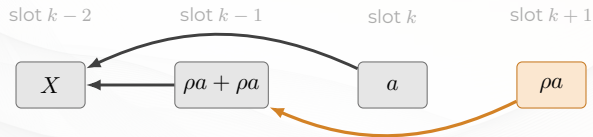
Ulysse Pavloff, Yackolley Amoussou-Guenou, Sara Tucci-Piergiovani

BACKUP: bouncing with obedient attesters



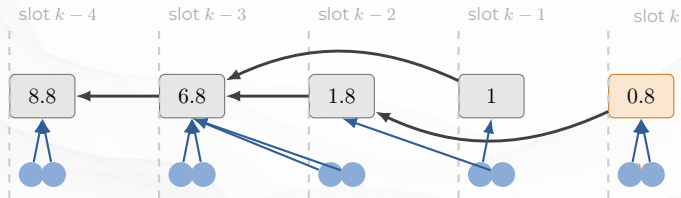
The proposer of slot k uses the boost to become head.

BACKUP: bouncing with obedient attesters



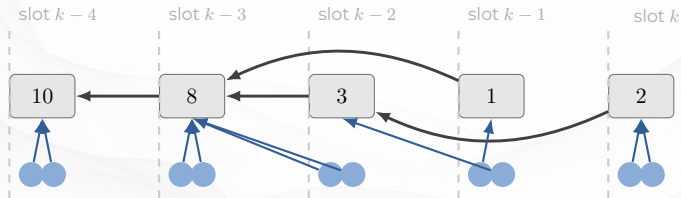
The proposer of slot $k+1$ can bounce back if $2\rho a \geq a$, i.e., $\rho \geq 1/2$.

BACKUP: fork choice with proposer boost



At the beginning of slot k , the timely block receives temporary proposer boost.

BACKUP: fork choice with proposer boost



At the end of slot k , the boost is cleared and only real attestations remain.

Backup: Strategic regimes as a function of proposer boost

Case	What can happen	Main implication
$\rho < \frac{1}{2}$	At most the first proposer can deviate. No sustained bouncing.	After one strategic fork, attestations create a gap larger than the proposer boost. Forking stops.
$\rho \geq \frac{1}{2}$ + obedient attesters	Bouncing may occur between consecutive proposers.	This is possible only if it is also profitable , i.e. (tx-fee condition is satisfied.)
$\rho \geq \frac{1}{2}$ + cunning attesters	Cunning attesters may anticipate the next proposer and stop supporting the current forking proposer.	This can deter proposer deviation : a bigger proposer boost can help restore obedience.